

$$\begin{matrix} \lambda & k \\ e^{\lambda} & k! \end{matrix}$$

$$X \sim P[X=k] = A \frac{[\ln(2)]^k}{k!}$$

$$k=0,1,\dots$$

$$\sum_{k=0}^{+\infty} A \frac{\lambda^k}{k!} = A \sum_{k=0}^{+\infty} \frac{\lambda^k}{k!} = A e^{\lambda} = 1 \rightarrow A = \frac{1}{e^{\lambda}} = \frac{1}{e^{\ln(2)}} = \frac{1}{2}$$

$$P[X=k] = P[Z=k | H_0]$$

$$\lambda = \ln(2)$$

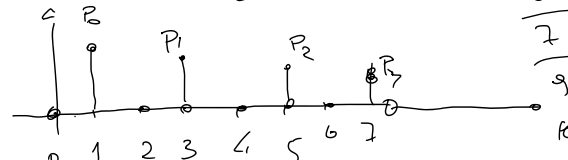
$$Y = 2X + 1$$



$$P[Y=k] = \begin{cases} 0 & k \text{ pari} \\ P[2X+1=k] = P[X = \frac{k-1}{2}] = \frac{1}{2} \frac{(\ln(2))^{(k-1)/2}}{((k-1)/2)!} & k \text{ dispari} \end{cases}$$

k	(k-1)/2
1	0
3	1
5	2
7	3
9	4

$$Z = \begin{cases} P[Z=k] = P[X=k] & \text{ipotesi } H_0 \\ P[Z=k] = P[Y=k] & \text{ipotesi } H_1 \end{cases}$$



$$P[H_0] = P[H_1] = \frac{1}{2}$$

$$P[Z=k | H_0] \begin{matrix} > \\ < \end{matrix} P[Z=k | H_1] = P[Z_{H_1}=k] = P[Y=k]$$

$$P[Z_{H_0}=k] = P[X=k]$$

$$P[X=k] \begin{matrix} > \\ < \end{matrix} P[Y=k]$$

$$P_k \begin{matrix} > \\ < \end{matrix} P_{\frac{k-1}{2}}$$

$$k \rightarrow \boxed{DEC} \rightarrow D(k)$$

$$D(k) = \begin{cases} 0 & k \text{ pari} \\ 1 & k \text{ dispari} \end{cases}$$

$$P(E) = P(H_0) P(E|H_0) + P(H_1) P(E|H_1)$$

$$P(E|H_1) = P[D=0|H_1] = P[Z \text{ sia pari} | H_1] = P[Z_{H_1} \text{ sia pari}] = P[Y \text{ sia pari}] = \sum_{k=0}^{+\infty} P[Y=2k] = 0$$

$$P(E|H_0) = P[D=1|H_0] = P[Z \text{ sia dispari} | H_0] = P[Z_{H_0} \text{ sia dispari}] = P[X \text{ sia dispari}]$$

$$\sum_{k=0}^{+\infty} \frac{(\ln 2)^{2k+1}}{(2k+1)!} = \frac{1}{2} \left[e^{\ln 2} - e^{-\ln 2} \right] = \sum_{k=0}^{+\infty} P[X=2k+1]$$

$$= \sum_{k=0}^{+\infty} \frac{1}{2} \frac{(\ln 2)^{2k+1}}{(2k+1)!} = \frac{1}{2} \sum_{k \text{ dispari}} \frac{\lambda^k}{k!} = \frac{1}{4} \left[e^{\lambda} - e^{-\lambda} \right] = \sum_{k=0}^{+\infty} P[X=2k+1]$$

k	2k+1
0	1
1	3
2	5
3	7

$$\sum_{k \text{ dispari}} \frac{\lambda^k}{k!} = \frac{1}{2} \left[\sum_{k=0}^{+\infty} \frac{\lambda^k}{k!} - \sum_{k=0}^{+\infty} \frac{(-\lambda)^k}{k!} \right] = \frac{1}{2} \left[e^{\lambda} - e^{-\lambda} \right] = \sinh(\lambda)$$

$$k \text{ pari} \rightarrow \lambda^k = (-\lambda)^k$$

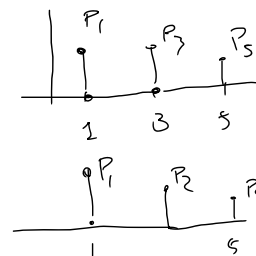
$$k \text{ dispari} \rightarrow \lambda^k = -(-\lambda)^k$$

$$P(E|H_0) = \frac{1}{4} \left[e^{\ln 2} - e^{\ln(1/2)} \right] = \frac{1}{4} \left[2 - \frac{1}{2} \right] = \frac{1}{4} \cdot \frac{3}{2} = \frac{3}{8}$$

$$P(E) = P(H_0) P(E|H_0) = \frac{3}{16}$$

$$Y = 2X - 1$$

$$P[Y=k] = P[2X-1=k] = P\left[X = \frac{k+1}{2}\right]$$



k	$\frac{k+1}{2}$
1	1
3	2
5	3

$$D = \begin{cases} 0 & k \text{ pari} \\ 0 & k=1 \\ 1 & k \text{ dispari, maggiore o uguale a 3} \end{cases}$$

$$P(E|H_1) = P(D=0|H_1) = P[(Z \text{ pari}) \text{ or } (Z=1)|H_1]$$

$$= P[Z \text{ pari} | H_1] + P[Z=1 | H_1] =$$

$$= P[Y \text{ pari}] + P[Y=1] = 0 + P_1 = P_1 = P_{2k+1} |_{k=0}$$

$$P(E|H_0) = P[D=1|H_0] = P[Z \text{ dispari e maggior o uguale di 3} | H_0]$$

$$= P[X \text{ sia dispari e maggior o uguale di 3}] = P_3 + P_5 + P_7 + \dots = \sum_{k=1}^{+\infty} P_{2k+1}$$

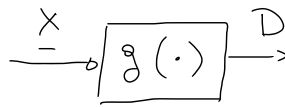
$$P(E) = P(H_0) \cdot P(E|H_0) + P(H_1) \cdot P(E|H_1)$$

$$= \frac{1}{2} (P_3 + P_5 + P_7 + \dots) + \frac{1}{2} P_1 = \frac{1}{2} (P_1 + P_3 + P_5 + P_7 + \dots) = \frac{1}{2} \sum_{k \text{ dispari}} P_k = \frac{3}{16}$$

$$\frac{X}{\sqrt{g(\cdot)}} \Big| \xrightarrow{D}$$

$$\frac{X}{\sqrt{g(\cdot)}} \Big| \xrightarrow{D}$$

$$D \in \{0, 1, \dots, K-1\}$$



$$D \in \{0, 1, \dots, K-1\}$$

$$\{x_1, x_2, \dots, x_n\}$$

$$\{p_1, p_2, \dots, p_n\}$$

$$\{q_1, q_2, \dots, q_n\}$$

$$f_X(x | H_i)$$

$$X | H_i \begin{cases} \{x_1, x_2, \dots, x_L\} \\ \{p_1^{(i)}, p_2^{(i)}, \dots, p_L^{(i)}\} \end{cases}$$

$$P(E | X=k)$$

$$P(\bar{E} | X=k) = P(H_0) \underbrace{P(\bar{E} | X=k, H_0)}_{=1} + P(H_1) \underbrace{P(\bar{E} | X=k, H_1)}_{=0}$$

$$g(x) = \begin{cases} 0 & x=k \end{cases}$$

$$P(E) = P(H_0) P(E | H_0) + P(H_1) P(E | H_1)$$

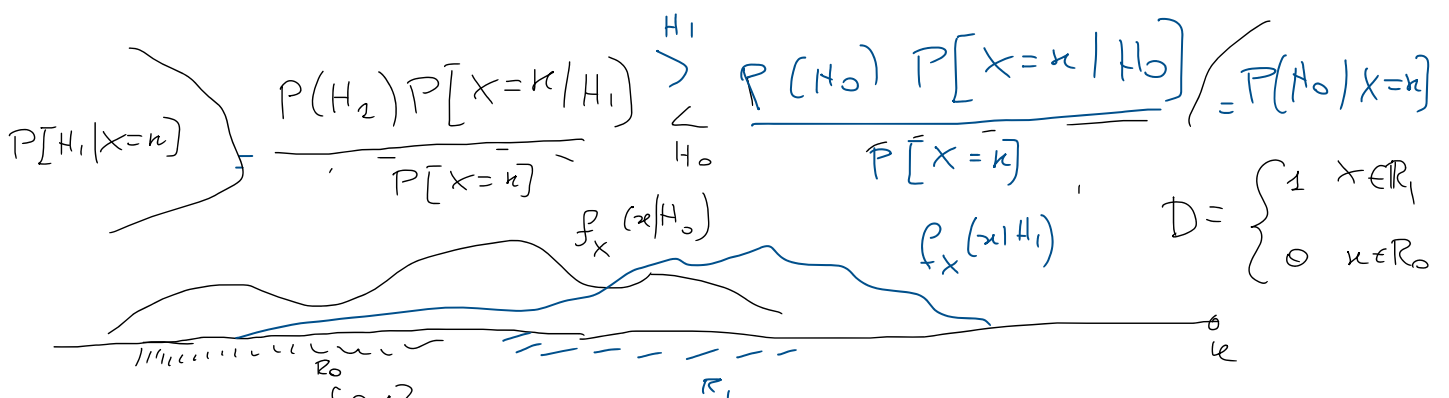
$$g(x) = \begin{cases} 1 & x=k \end{cases}$$

$$P(E | H_0) = P[X \in R_1 | H_0] = P[X | H_0 \in R_1]$$

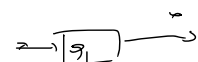
$$P(E | H_1) = P[X | H_1 \in R_0]$$

$$P(E | H_0) = \sum_{k \in R_1} P(H_0) P[X=k | H_0]$$

$$P(E | H_1) = \sum_{k \in R_0} P(H_1) P[X=k | H_1]$$



$$P(e) = \int_{-\infty}^{+\infty} \underbrace{f_X(x)}_{\geq 0} \underbrace{P(e | X=x)}_{\geq 0} dx$$



$$P(e | X=x) = P(H_0) P(e | H_0, X=x) + P(H_1) P(e | H_1, X=x)$$

$$P(C|X=x) = P(H_0) \underbrace{P(C|H_0, X=x)}_{=1} + P(H_1) \underbrace{P(C|H_1, X=x)}_{=0} = \boxed{1}$$

$$P(C|X=x) f_X(x) = \int_{X|C} f_X(x) P(C)$$

$$P(E) = P(H_0) P(E|H_0) + P(H_1) P(E|H_1)$$

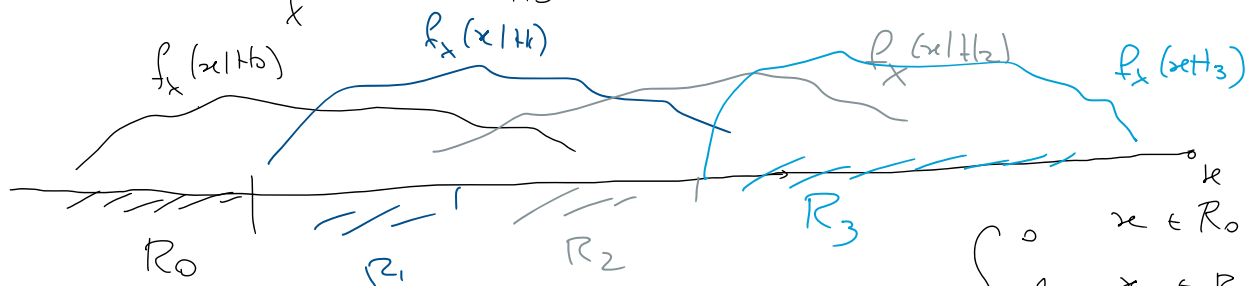
$$= P(H_0) P(X \in R_1 | H_0) + P(H_1) P(X \in R_0 | H_1)$$

$$= P(H_0) \int_{R_1} f_{X|H_0}(x) dx + P(H_1) \int_R f_{X|H_1}(x) dx$$

$$= \int_{R_1} P(H_0) f_{X|H_0}(x) dx + \int_{R_0} P(H_1) f_{X|H_1}(x) dx$$

$$P(H_1) f_{X|H_1}(x) > P(H_0) f_{X|H_0}(x) \Rightarrow P(H_0|X=x) = \frac{P(H_0) f_{X|H_0}(x)}{P(H_0) f_{X|H_0}(x) + P(H_1) f_{X|H_1}(x)}$$

$$P(H_1|X=x) = \frac{P(H_1) f_{X|H_1}(x)}{f_X(x)} > \frac{P(H_0) f_{X|H_0}(x)}{f_X(x)} = P(H_0|X=x)$$



$$P(C) = P(H_0) P(C|H_0) + P(H_1) P(C|H_1) + P(H_2) P(C|H_2) + P(H_3) P(C|H_3)$$

$$D = \begin{cases} 0 & x \in R_0 \\ 1 & x \in R_1 \\ 2 & x \in R_2 \\ 3 & x \in R_3 \end{cases}$$

$$= P(H_0) P[X \in R_0 | H_0] + P(H_1) P[X \in R_1 | H_1] + P(H_2) P[X \in R_2 | H_2]$$

$$+ P(H_3) P[X \in R_3 | H_3] \quad C|_{H_0} \Rightarrow D=0 = X|_{H_0}$$



$$= P(H_0) \int_{R_0} f_x(x|H_0) dx + P(H_1) \int_{R_1} f_x(x|H_1) + P(H_2) \int_{R_2} f_x(x|H_2) + P(H_3) \int_{R_3} f_x(x|H_3)$$

$$D = \arg \max_{i \in \{0,1,2,3\}} P(H_i) f_x(x|H_i)$$

$$P(H_0) f_x(x|H_0)$$

$$P(H_1) f_x(x|H_1)$$

$$P(H_2) f_x(x|H_2)$$

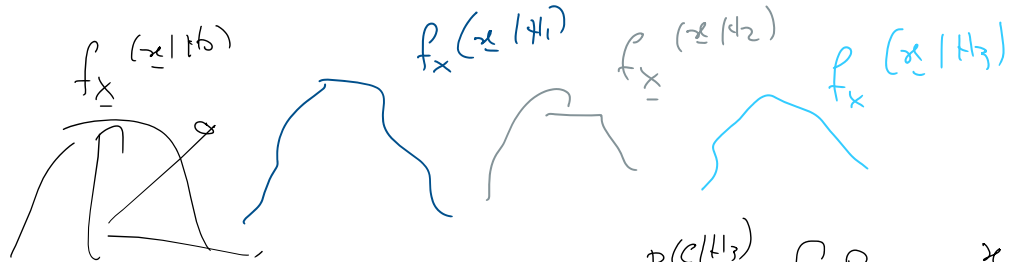
$$P(H_3) f_x(x|H_3)$$

$$P(H_0|x=x) = \frac{P(H_0) f_x(x|H_0)}{f_x(x)}$$

$$P(H_1|x=x) = \frac{P(H_1) f_x(x|H_1)}{f_x(x)}$$

$$P(H_2|x=x) = \frac{P(H_2) f_x(x|H_2)}{f_x(x)}$$

$$P(H_3|x=x) = \frac{P(H_3) f_x(x|H_3)}{f_x(x)}$$



$P(C)$

$$P(H_0) P(C|H_0) + P(H_1) P(C|H_1) + P(H_2) P(C|H_2) + P(H_3) P(C|H_3)$$

$$P[x \in R_0 | H_0]$$

$$\int_{R_0} f_x(x|H_0)$$

$$\int_{R_0} P(H_0) f_x(x|H_0) dx + \int_{R_1} P(H_1) f_x(x|H_1) dx + \dots + \int_{R_3} P(H_3) f_x(x|H_3) dx$$

$$D = \arg \max_{i \in \{0,1,2,3\}} P(H_i) f_x(x|H_i)$$

$$D = \begin{cases} 0 & x_i: i \in I_0 \\ 1 & x_i: i \in I_1 \\ \vdots & \vdots \end{cases}$$

$$c \in \{0, 1, 2, 3\}$$

$$c \in \{0, 1, 2, 3\}$$

$$\{x_1, x_2, \dots, x_n\}$$

$$\{p_1^{(c)}, p_2^{(c)}, \dots, p_n^{(c)}\} \quad c = 0, 1, 2, 3$$

$$D = \begin{cases} 1 & x_i : i \in I_1 \\ 2 & x_i : i \in I_2 \\ 3 & x_i : i \in I_3 \end{cases}$$

$$P(c) = P(H_0)P(c|H_0) + \dots + P(H_3)P(c|H_3)$$

$$= P(H_0)P[X=x_i \text{ con } i \in I_0 | H_0] + \dots + P(H_3)P[X=x_i \text{ con } i \in I_3 | H_3]$$

$$D = \underset{c \in \{0, 1, 2, 3\}}{\operatorname{argmax}} \underbrace{P(H_i)P[X=x_i | H_i]}_{\alpha} \xrightarrow{x_i} \boxed{g(\cdot)} \rightarrow D$$